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Towards the understanding of the unusual rheological response of polymer nanocomposites☆

Thaleia-Michaela Chatzaki [a,b], Sokratis Kogchylakis [a,b], Dimitris Vlassopoulos [a,c], Spiros H. Anastasiadis [a,b,*], Kiriaki Chrissopoulou [a,*]

[a] Institute of Electronic Structure and Laser, Foundation for Research and Technology-Hellas, N. Plastira 100, Vassilika Vouton, 700 13 Heraklion Crete, Greece
[b] Department of Chemistry, University of Crete, 710 03 Heraklion Crete, Greece
[c] Department of Materials Science and Technology, University of Crete, 700 13 Heraklion Crete, Greece

ABSTRACT

It is well known that the properties of polymers can be improved by the incorporation of nano-additives in the polymeric matrix, forming nanocomposite materials that often possess innovative physicochemical properties compared to the initial materials and / or the respective conventionally filled systems. In this work, we investigate the rheological response of different series of poly(ethylene oxide) / silica, PEO/SiO_2, nanocomposites by shear rheology. More specifically, spherical SiO_2 nanoparticles of two different radii are dispersed in PEOs of different molecular weights covering the regime from unentangled to highly-entangled systems. Nanocomposites over a wide range of compositions were developed in order to investigate both the effect of the size and concentration of the nanoparticles and the effect of the molecular weight of the polymer matrix, on the rheological properties of the nanocomposite materials. For all systems three different concentration regimes are observed with distinctly different rheological response; a 'polymer-like' regime, where the behavior resembles the one of the respective polymers, a 'weak gel-like' regime at higher nanoparticle content, where enhanced polymer / particle interactions modify the behavior, and a 'strong gel-like' one characterized by percolated networks of nanoparticles formed within the polymer matrices. The 'polymer-like' regime is especially interesting since a significant reduction of the melt viscosity is observed, which is attributed to a dilution of the entanglement network of the polymer chains due to the presence of the nanoparticles.

1. Introduction

Polymer nanohybrids, composed of a polymer matrix and inorganic or carbon nanoadditives (for example, nanoparticles, nanotubes, layered materials like clays and/or graphitic nanomaterials, etc.) as the nanofiller, possess improved and often innovative physicochemical properties compared to either the neat polymer matrices or their conventionally filled systems [1–18]. The investigation of the structure, chain conformation and dynamics in polymer nanocomposites constitutes a very active research area since their understanding is very important for the utilization of these materials in a broad range of potential or current applications [19–33].

Poly(ethylene oxide), PEO, is a non-ionic, water soluble, semi-crystalline polymer that is widely utilized in various applications mainly due to its hydrophilicity, biocompatibility and its ability for ion conduction [11,34–37]. PEO possesses a significantly high degree of crystallinity, which, however, prevents ion transport at room temperature and, therefore, a number of investigations aimed at increasing its amorphous volume fraction and, thus, improving conductivity [38,39]. One promising approach towards decreasing PEO crystallinity and, at the same time, improving its mechanical properties is by incorporating inorganic nanoadditives to obtain nanocomposites [20,26,29,40,41]; it has been shown that the chain confinement as well as the adsorption capacity of the nanoadditive surfaces are the main parameters that define the crystallinity reduction. Among the various nanoadditives, silica (SiO_2) nanoparticles have been utilized to develop PEO / SiO_2 nanocomposites; the polymer morphology, chain conformation as well as their dynamics have been investigated and the effect of the presence of the additives and the confinement they induce to the polymer chains have been investigated [28,29,33]. In this system, attractive interactions exist between the poly(ethylene oxide) and the silica surfaces, mainly due to the formation of hydrogen bonds, that leads polymer chains to adsorb onto the nanoparticle surface stabilizing the system [28,29,42–45].

Among the different properties of polymer nanocomposites, the understanding of their rheological and/or mechanical properties is

☆ This manuscript is dedicated to Professor Nikos Hadjichristidis on the occasion of his 80th birthday.

* Corresponding authors at: Institute of Electronic Structure and Laser, Foundation for Research and Technology-Hellas, N. Plastira 100, Vassilika Vouton, 700 13 Heraklion Crete, Greece (S.H. Anastasiadis, K. Chrissopoulou).

E-mail addresses: spiros@iesl.forth.gr (S.H. Anastasiadis), kiki@iesl.forth.gr (K. Chrissopoulou).

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considered vital since they define the processability of the materials as well as their potential utilization; the investigation of these properties for nanohybrids containing spherical nanoparticles is a very active area of research. However, despite the anticipated and widely reported effect of the reinforcement of the polymer matrix with the incorporation of nanoparticles [46–49], the mechanism linking this macroscopic effect with the complex microscopic dynamics of the polymer chains and the nanoparticles is still not fully understood.

In general, there seems to be a consensus that the state of dispersion of the nanoparticles within the polymer matrix can crucially affect the rheological and the mechanical properties of the nanocomposites. When nanoparticle aggregates are formed, the mechanical properties of the material will be dominated by the clusters, with some works in the literature attributing the mechanical reinforcement purely to such particle superstructures that span the entire system [50,51]. However, even in the case of favorable nanoparticle-polymer interactions, which result in the development of well-dispersed nanohybrids, the rheological properties of the hybrid material are remarkably different in comparison to the respective ones of the bulk polymer [48,52,53,54]. In this case, these favorable interactions facilitate the adsorption of the polymer on the nanoparticle surface, often mentioned as the "bound layer", which exhibits reduced segmental mobility. Other studies suggest the existence of a glassy (or dynamically frozen) bound layer surrounding the nanoparticles, attributing the mechanical reinforcement of the polymer nanocomposites to the percolation of overlapping glassy polymer bound layers at high nanoparticle loadings [55,56]. While the existence of this bound layer has been confirmed by other works in the literature, it is also suggested that it does not have a glassy nature but, on the contrary, it is highly mobile [57]. In a more recent work, it was suggested that a network is formed with the nanoparticles acting as the nodes linked by flexible polymer strands, which transform into glassy bridges at a critical nanoparticle volume fraction, once the interparticle distance approaches the Kuhn length of the chains (~1 nm) [58]. A recent computational work attributes the mechanical reinforcement to a two-fold mechanism, suggesting that, at low nanoparticle loadings, it is due to the higher rigidity of the polymer/nanoparticle interphase as compared to that of the bulk, whereas, at the higher nanoparticle loading regime, the mechanical enhancement is augmented due to the coupling between the train and bridge conformations of the adsorbed polymer chains [59].

In most cases, a reinforcement of the mechanical and/or rheological properties of polymer nanocomposites is reported when nanoparticles are introduced into the polymer matrix; however, there have been few works that discuss a non-Einstein-like decrease in the viscosity, also known as fluidification, that was induced by the incorporation of nanoparticles into the polymer matrix. Such a behavior, if its origin is understood, can be of great industrial interest, since it could lead to easier polymer processing at lower temperatures and shorter times. A reduction in viscosity in PVC (nano)composites has been first reported as early as in the 1970s when, with the addition of nanoparticles, the polymer melt viscosity decreased, presumably due to the creation of additional free volume near the boundary surface region between the polymer and the nanoparticles [60]. Since then, there have been a few more works reporting a weak reduction of viscosity; however, there has been no clear understanding on the origin of this behavior [61–69]. In one of them, polystyrene nanoparticles were dispersed in linear polystyrene matrices, therefore eliminating enthalpic interactions, serving as an ideal model system [61]; a decrease of the zero shear viscosity was observed, that was constant for the different concentrations examined, and was attributed to the enhancement of free volume. In a similar system, comprising single chain polystyrene nanoparticles within linear polystyrene matrices, both simulations and rheological measurements [67] showed a reduction of viscosity of the nanocomposite melt, with the effect being larger when high molecular weight polymer matrices were utilized; these results were explained via a reduced friction experienced by polymer segments that are comparable in size and in direct interaction with the single chain polystyrene nanoparticles. In one of the recent works in the literature [70], this phenomenon is studied using Au nanoparticles coated with short PEO chains (molecular weight 1000 g/mol) within an entangled PEO matrix; it was observed that the small nanoparticles cause a significant viscosity reduction whereas larger nanoparticles cause mechanical reinforcement, with the former being attributed to disentanglement of the polymer chains rather than plasticization of the polymer. Despite the many reports of this very interesting fluidification phenomenon and the plethora of different hypotheses attempting to explain the behavior, understanding of the exact mechanism of its origin remains an unanswered question.

In this work, we provide solid experimental evidence towards advancing the understanding of the unusual rheological behavior of well-dispersed polymer nanocomposites by investigating poly(ethylene oxide)/silica, PEO/SiO_2, hybrids. Series of nanocomposites with varying compositions utilizing PEO chains of different molecular weights and SiO_2 nanoparticles of different sizes were developed. A highly-entangled, a marginally entangled and an unentangled polymer matrix were utilized. At the same time, the radii of gyration, R_g, of these chains are larger, comparable and smaller in size than the small nanoparticles (NP_7) and much smaller when compared to the larger nanoparticles (NP_{67}) that were utilized. We demonstrate that in all systems, at low nanoparticle loadings, the dynamics of the nanocomposite materials resemble those of the respective neat polymers. Above a critical nanoparticle loading, a change in the behavior occurs indicating enhanced polymer/particle interactions that result to 'weak gel-like' dynamics; however, the temperature dependence is still Arrhenius with only a slight increase in the activation energy. At higher loadings, a percolated nanoparticle network is formed leading to a 'strong gel-like' behavior, accompanied by a significant increase in the activation energy, which remains Arrhenius-like for all nanoparticle concentrations. The effects of both the nanoparticle size and concentration as well as the polymer molecular weight on the behavior are investigated; the three concentration regimes are observed in all nanocomposite series, however, at different nanoparticle loadings depending on the system. A remarkable reduction of the viscosity is observed for both nanoparticle sizes, for the nanocomposites that are in the 'polymer-like' regime (low nanoparticle loading); this reduction reaches almost two orders of magnitude for the polymer with the high molecular weight that is considered well-entangled, it decreases in intensity for marginally entangled PEO chains, whereas it is reversed in the case of the unentangled PEO chains, for which the viscosity values show a monotonous increase. It is the first time that a direct connection between the entanglements (or M_w) and the behavior of the viscosity of the polymer nanocomposites is clearly shown attributing the significant decrease of the complex viscosity observed at low nanoparticle content to the disentanglement of the polymer chains.

2. Materials and methods

2.1. Materials

The poly(ethylene oxide), PEO, homopolymers that were utilized in this work were purchased from Sigma Aldrich. Several different molecular weights (M_w) were studied ranging from highly to marginally entangled to unentangled PEO; their characteristics are listed in Table 1.

Two different silica (SiO_2) nanoparticles were utilized, that were purchased from Sigma Aldrich (Ludox LS) and Nissan Chemical (Snowtex ZL), in aqueous dispersions 30 % wt and 40 % wt, respectively. The radii of the nanoparticles were measured with Dynamic Light Scattering (DLS) and via Transmission Electron Microscopy (TEM) and are shown in Table 1. Both techniques indicate that the nanoparticles are fairly monodispersed and illustrate the absence of aggregation in dispersion.

The silica nanoparticles possess hydroxyl (–OH) groups on their surfaces. One can calculate their density on the nanoparticle surface by utilizing Thermogravimetric Analysis (TGA) data (Figure S1 of the

Table 1
Molecular Characteristics of the PEO Homopolymers and the Silica Nanoparticles.

Code	M_w (Kg/mol)	M_w/M_n *	Hydrodynamic Radius† (nm)	Radius of Gyration# (nm)	Nanoparticle Size (nm)
PEO300k	300,000	2.4	36.2 ± 5	46 ± 5	–
PEO100k	100,000	2.4	12.0 ± 2	15 ± 2	–
PEO35k	35,000	1.07	7.1 ± 1	9 ± 1	–
PEO8k	8,000	1.04	3.0 ± 0.4	3.8 ± 0.4	–
PEO3.35 k	3,350	1.05	1.9 ± 0.3	2.4 ± 0.3	–
NP_7	–	–	–	–	7 ± 1 (DLS)& 6 ± 2 (TEM)§
NP_{67}	–	–	–	–	67 ± 1 (DLS)& 68 ± 2 (TEM)§

* determined by size exclusion chromatography.
† hydrodynamic radius measured utilizing Dynamic Light Scattering (DLS) at T = 25 °C in water.
# calculated based on the hydrodynamic radii measured utilizing Dynamic Light Scattering (DLS) assuming Θ-solvent conditions.
& nanoparticle radius measured utilizing Dynamic Light Scattering (DLS).
§ nanoparticle radius measured utilizing Transmission Electron Microscopy (TEM).

Supplementary Material). More specifically, the mass loss, ΔM, between 200 °C – 800 °C, has been attributed to the loss of the –OH groups on the particle surface; for NP_7 this is ~ 0.02 g/g whereas for NP_{67} this is 0.015 g/g of nanoparticle. The –OH density is calculated using the equation $D = \Delta M/_{S \times W_{OH}} N_{AVOG}$ [71], where D is the OH density (nm^{-2}), ΔM is the mass loss per mass of silica, S is the surface area of silica (area per mass), W_{OH} is the molar mass of OH and N_{AVOG} is Avogadro number. Estimating the nanoparticle surface area either utilizing the measured nanoparticle radius or the materials data sheets results in –OH densities on the nanoparticles in the range of D = 2.8–4.4 OH/nm^2 for NP_7 and D = 15.6–31.6 OH/nm^2 for NP_{67}.

Nanocomposite Preparation: PEO/SiO_2 nanocomposites were prepared via solution mixing over a range of compositions. Initially, a PEO/water solution was prepared and, then, the appropriate quantity of the aqueous dispersion of the nanoparticles was added. The systems were left to mix for 24 h, were transferred to a Petri dish and dried under vacuum. Finally, following the solvent evaporation, the samples were annealed at T = 100 °C in vacuum, to ensure equilibration.

The concentration of the nanoparticles was confirmed using Thermogravimetric Analysis (Figure S2 of the Supplementary Material); these measurements show that the decomposition temperature of the polymers remains relatively constant for all nanocomposites, as well.

2.2. Rheology

The rheological measurements for PEO300k and PEO100k were performed using the RMS-800 strain controlled rheometer (former Rheometrics, USA), while a MCR-702 twin drive rheometer (Anton-Paar, Austria) was utilized for the measurements of the PEO35k, PEO8k, and PEO3.35 k systems; in both cases, a 25 mm stainless steel parallel plate geometry was used with a sample thickness in the range 0.7–1 mm. In both cases, temperature control was achieved by means of convection oven, operating with nitrogen gas to create an inert atmosphere. Samples were loaded at temperature of 80 °C, which is higher than the melting temperature of the investigated polymers. Following the loading of the pellets, the equilibration of the samples was recorded via consecutive dynamic frequency sweep tests (DFS) in the linear viscoelastic regime (LVE); the latter has been determined by an initial dynamic strain sweep test (DSS). In all cases, the rheological measurements were performed once the samples had reached equilibrium (determined by the steady state response), as illustrated in Figure S3 of the Supplementary Material. Under these conditions, the LVE regime of each sample was determined by DSS tests (for every temperature) and followed by dynamic frequency sweep tests (DFS) measurements over a frequency range of 100–0.01 rad/s. The temperature range 60 °C – 120 °C was explored, with the lower one being restricted by the polymer crystallization temperature, T_c. The rheological response of the polymers with the lower molecular weights (PEO8k and PEO3.35 k), as well as of some of their nanocomposites that behaved as viscous liquids, was probed with steady shear measurements, by performing step rate (SR) tests at different shear rates. Each test was followed by a waiting period of 180 s, allowing stress to fully relax before the next SR test. It is noted that no evidence of edge fracture was observed during the SR measurements (based on visual observations and the reached steady state value of the shear stress growth coefficient).

2.3. Transmission Electron Microscopy

The size of the two nanoparticles as well as the degree of dispersion of the nanoparticles in the polymer matrix was investigated by utilizing a JEOL JEM 2100. For the nanoparticle images, the initial dispersions in water (30 wt% for the NP_7 and 40 wt% for the NP_{67}) were diluted 10 times, and, then, they were placed onto the TEM grids. For the nanocomposites, the dispersions were diluted depending on the concentration of the nanoparticles before being placed onto the TEM grids; for all samples an accelerating voltage at 200 kV was used. The contrast between the silica-containing regions (shown as dark spheres) and the polymer (bright region) was sufficient for imaging, and no further staining was required.

3. Results and discussion

Fig. 1a shows master curves for the three neat poly(ethylene oxide) polymers with molecular weights 35 k, 100 k and 300 k. They were obtained from DFS measurements at different temperatures (60–120 °C) and application of time temperature superposition (TTS) principle at a reference temperature T_{ref} = 80 °C. It is clear that in all cases the materials are thermo-rheologically simple. Moreover, in all cases and at low frequencies, the values of the storage modulus, G', are lower than those of the loss modulus, G″. More specifically, for the PEO35k, G' and G″ follow the terminal scaling with ω^2 and ω^1 dependence, respectively. For the polymers with larger molecular weight, the flow regime is not clearly attained due to the emergence of slower relaxation modes and the (associated) higher polydispersity; it is clear that the longest relaxation time for the PEO35k is in the high frequency side and well outside the measurement window, whereas it can be observed for the PEO100k and it becomes slower for the PEO300k, as anticipated. At the same time, both the storage and the loss moduli show a dependence on the PEO molecular weight. The shift factors, a_T, extracted from the TTS follow Arrhenius temperature dependencies (shown in Fig. 2c, below, for one of the homopolymers, the PEO35k, and its nanocomposites), as expected by the fact that the measurements are performed at temperatures well above the glass transition temperature of the polymers. The respective activation energies are $E_{a,PEO35k}$ = 29.1 kJ/mol, $E_{a,PEO100k}$ = 33.2 kJ/mol and $E_{a,PEO300k}$ = 38.8 kJ/mol, exhibiting a dependence on the PEO molecular weight.

Fig. 1b shows the molecular weight dependence of the zero shear viscosity for all polymers at T = 80 °C. Due to the large difference in the molecular weights and therefore in the viscosity of the polymers a different approach was followed for the determination in each case. For PEO35k, the zero-shear viscosity can be obtained directly from the DFS measurements. For PEO100k and PEO300k, since the flow regime could not be clearly reached even at the highest temperature measured, the empirical Cross model was utilized to fit the data. It describes the viscosity $\eta = \eta_0 / \left[1 + (\lambda\dot{\gamma})^{(1-n)}\right]$, where η_0 is the zero-shear viscosity, $\dot{\gamma}$ is the

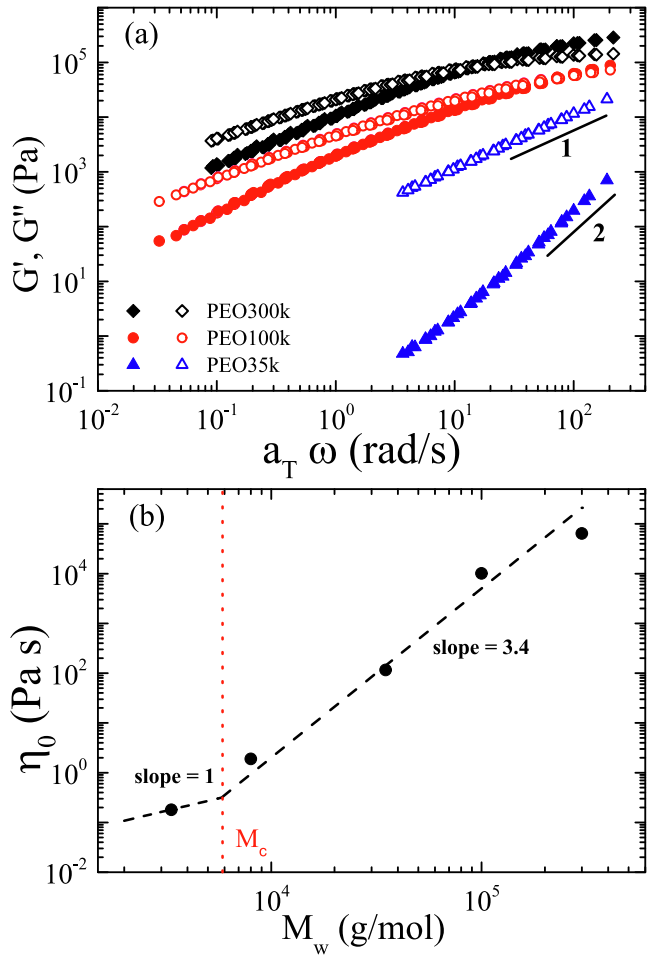

Fig. 1. (a) Frequency dependence of the storage, G', (solid symbols) and the loss, G″, (open symbols) moduli for PEO35k, PEO100k and PEO300k presented following time–temperature superposition and (b) zero shear viscosity as a function of the PEO molecular weight measured at T = 80 °C. The dotted lines indicate the bibliographical M_c for PEO [76–78] and the dash lines indicating the slopes of 1 and 3.4 are to guide the eye. It is noted that the error is smaller than the size of the points. The data in (a) are shifted along the horizontal axis keeping 80 °C as the reference temperature.

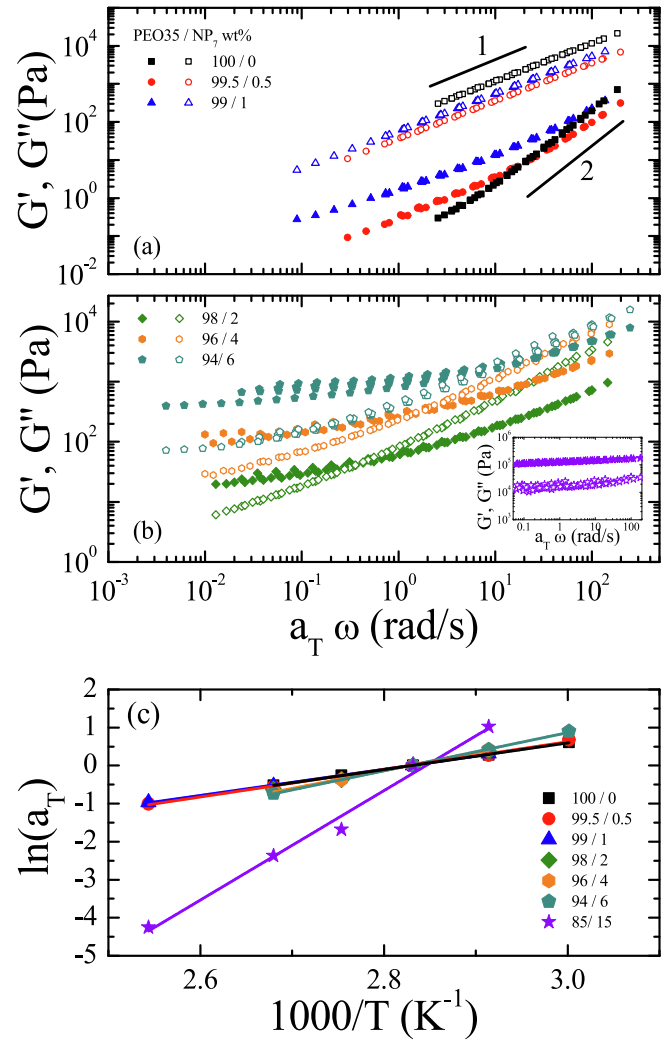

Fig. 2. Frequency dependence of the storage, G' (solid symbols), and the loss, G″ (open symbols), moduli for (a) the PEO35k and nanocomposites with 0.5 and 1 wt% NP_7 and (b) nanocomposites with 2, 4 and 6 wt% NP_7 following time–temperature superposition principle at a temperature range 60 to 120 °C. The data in both (a) and (b) are shifted along the horizontal axis keeping 80 °C as the reference temperature. The inset in (b) illustrates the two moduli for a nanocomposite with 15 wt% NP_7. (c) Temperature dependence of the shift factors a_T in an Arrhenius representation.

shear rate, λ is the relaxation time and n is the power law index [72]; Figure S4 of the Supplementary Material shows the quality of the corresponding fits for PEO100k and PEO300k. On the other hand, for PEO8k and PEO3.35k, the viscosity was measured with steady shear measurements as already mentioned (see also below). It is evident that the viscosity increases with the expected scaling behavior; indeed, taking into account the literature values for the entanglement molecular weight of PEO, M_e, (between 1620 and 2220 g/mol) [71,72,73] , the obtained viscosity values conform to the $\eta \propto MW^1$, in the unentangled, and $\eta \propto MW^{3.4}$ in the entangled regime, respectively, with the transition occurring at the critical molecular weight, M_c, that is 2.2 times the M_e [73]. Therefore PEO3.35k can be considered as unentangled, PEO8k as slightly entangled and PEO35k, PEO100k and PEO300k as well entangled.

The rheological behavior of the polymer nanocomposites was investigated as well. Fig. 2 shows the storage and loss moduli of a series of PEO/SiO_2 nanohybrids composed of PEO35k and NP_7 for different concentrations; in all cases the data were shifted along the horizontal axis keeping the temperature of 80 °C as the reference temperature. It is clear that a good superposition of the data for the different temperatures can be obtained for most compositions. As discussed above, the neat polymer is clearly in the flow regime as the storage, G', and loss moduli, G″, show the predicted ω^2 and ω^1 dependencies, respectively (Fig. 2a). For the nanocomposites with relatively low nanoparticle content, i.e., for 0.5 and 1 wt% NP_7, the behavior is very similar to the neat polymers within the accessible frequency range. The nanohybrids exhibit a liquid-like behavior with the values of the G' being lower than the ones of the G″. The loss modulus G″ preserves the $\sim \omega^1$ dependence; however, there is a tendency for the data of G' to display a smaller slope at lower frequencies (Fig. 2a), which is not due to any phase angle resolution issues. Increasing the nanohybrid concentration (for NP_7 content between 2 and 6 wt%), the observed behavior changes with both moduli approaching a plateau at low frequencies (Fig. 2b) indicating a 'weak gel-like' response [74]. This results in the observation of a crossing of G' and G″, and therefore in the observation of a characteristic time; this characteristic time becomes faster with increasing NP_7 content. It is noted that even at longer time scales, obtained via Creep measurements of these systems (Figure S5 of the Supplementary Material), no terminal flow is observed, with the slope being significantly lower than 1. Further increase of the nanoparticle composition leads to an additional change in the behavior, where a 'strong gel-like' behavior is observed (for 15 wt

% NP_7), with the values of G' being almost one order of magnitude larger than the ones of G″, whereas both moduli do not show any frequency dependence (inset of Fig. 2b). As we described above, TTS holds for the majority of the concentrations discussed in Fig. 2. For the nanohybrid with 6 wt% NP7, however, one can observe the beginning of a failure of TTS especially for the lower frequencies; this can be associated with the presence of an additional dynamic process that is not correlated with the microscopic monomeric friction and can possibly be attributed to an adsorption/desorption mechanism of the polymer chains on the nanoparticles [75].

Since TTS holds for most of the nanohybrid compositions, the temperature dependence of the shift factor a_T can be examined (Fig. 2c). An apparent Arrhenius dependence is observed over the specific temperature range; however, there is a systematic change of the activation energy for the three different composition regimes that were described above. For the two lower nanoparticle concentrations where a clear polymer terminal response is observed, the activation energy is very similar to the one of the neat polymer (30.6 and 28.6 kJ/mol for the nanohybrids with 0.5 and 1 wt% of NP_7, respectively, versus 29.1 kJ/mol for the neat PEO35k). For the intermediate concentration regime, an increase of E_a is observed with values 35.7, 39.1 and 40.7 kJ/mol for the hybrids with 2, 4 and 6 wt% of NP_7, respectively. Finally, for the nanocomposite with the larger concentration of nanoparticles (i.e., 15 wt%), where a 'strong gel-like' behavior is observed, a significant increase in the activation energy is observed, which becomes $E_A = 119.7$ kJ/mol, illustrating enhanced difficulty for large-scale thermal motion.

To understand the behavior and correlate the obtained rheological response to the morphology and the state of dispersion of the nanoparticles within the polymer matrix, Transmission Electron Microscopy (TEM) images of nanocomposites in the different regimes have been taken. Fig. 3 shows representative TEM micrographs of nanohybrids with 2 wt% NP_7 (Fig. 3a) and 15 wt% NP_7 (Fig. 3b), respectively. In the former case, the presence of well-dispersed nanoparticles that are far apart from each other can be observed; this precludes the possibility that the change in the viscoelastic response of the polymer is due to spatial chain confinement between the nanoparticles. Alternatively, the 'weak gel-like' response can be attributed to the preferential adsorption of the PEO chains on the SiO_2 surfaces because of the favorable interactions between the oxygen groups of the chains and the hydroxyl functional groups on the nanoparticles. In an earlier work of ours on the dynamics of PEO/SiO_2 nanocomposites studied both experimentally and theoretically [33], the characteristic chain desorption time was calculated to vary between 4 and 80 ns over a temperature range of 400–330 K when 2 vol% (~4 wt%) of silica nanoparticles were incorporated within the PEO matrix.

On the other hand, when the content of the nanoparticles is increased to 15 wt%, it is clear by the TEM image (Fig. 3b) that a dense percolated network of the NP_7 nanoparticles is formed within the PEO matrix; the existence of this network gives rise to the 'strong gel-like' response observed in the rheological measurements. Moreover, it is emphasized that no aggregation is observed for neither the 2 wt% nor the 15 wt% concentration of nanoparticles. It also noted that the same behavior is observed for all nanocomposites with Fig. 3c and 3d illustrating this for the PEO3.35k with the lowest molecular weight studied in this work whereas micrographs for the PEO100k/NP7 were shown earlier [29].

Fig. 4 shows the frequency dependence of the storage and loss moduli of nanocomposites that contain the PEOs of the two high molecular weights, PEO100k and PEO300k, for different NP_7 concentrations, following time–temperature superposition. A similar rheological behavior with what was described for the PEO35k / NP_7 nanohybrids is observed. For both polymer nanocomposites series, on one hand, the TTS principle holds for the nanohybrids with low nanoparticle content (Fig. 4a and 4c). TTS starts to fail with increasing the nanoparticle concentration and especially for 10 wt% NP7 (Fig. 4b and 4d). On the other hand, the behavior for the three concentration regimes, i.e., the 'polymer-like', the 'weak gel-like' and the 'strong gel-like' ones, are observed. For PEO100k and for nanohybrids with 2 and 4 wt% NP_7 (Fig. 4a), the storage modulus, G', is lower than the loss modulus, G″, over nearly the entire frequency range; the two moduli cross at higher frequencies marking the terminal relaxation time of the polymer. It is

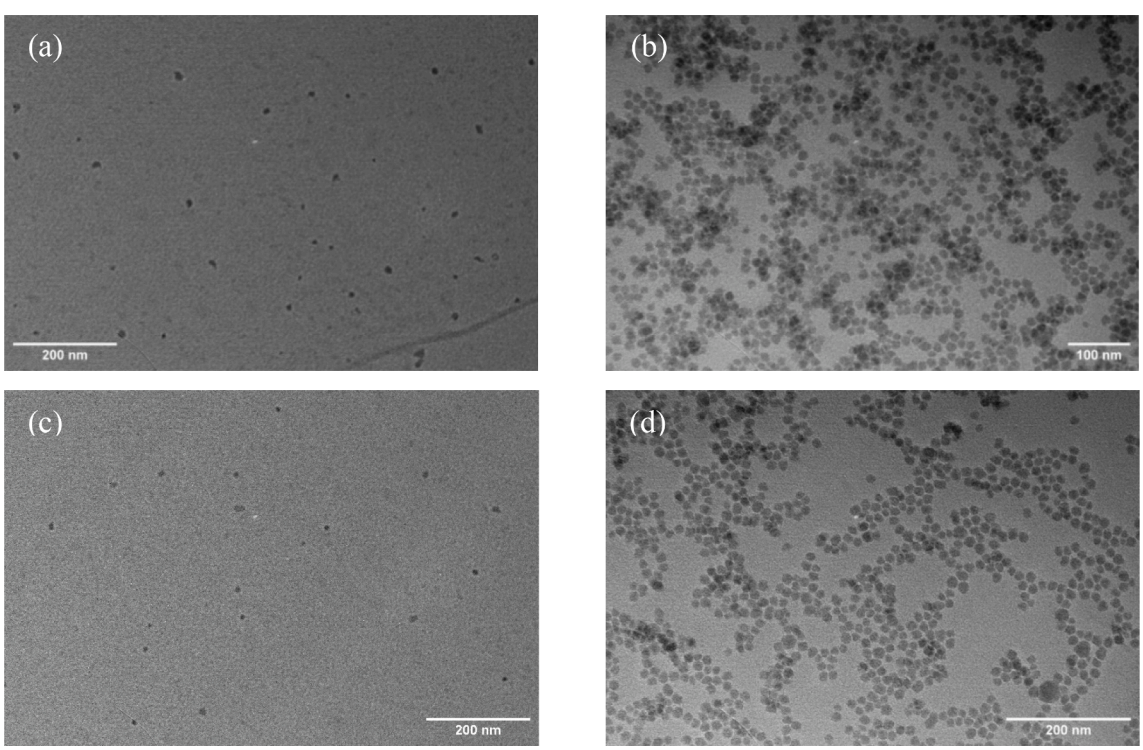

Fig. 3. Transmission Electron Microscopy (TEM) images of NP_7 nanocomposites with PEO35k (a,b) and PEO3.35k (c,d) with concentrations 2 wt% (a,c), 15 wt% (b) and 10 wt% (d).

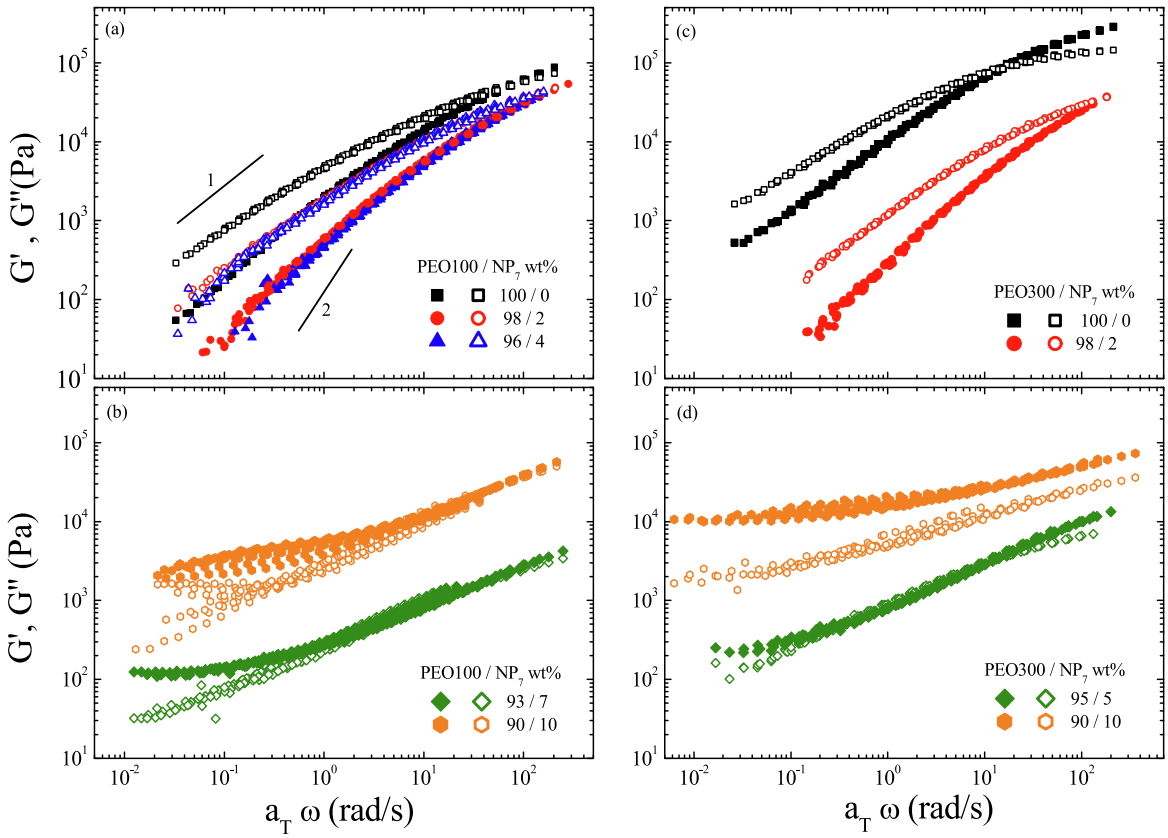

Fig. 4. Frequency dependence of the storage modulus, G' (solid symbols), and the loss modulus, G″ (open symbols), for PEO100k / NP_7 (a, b) and PEO300k / NP_7 (c, d) nanohybrids for different concentrations in the temperature range 60 to 120 °C. The data are shifted along the horizontal axis keeping 80 °C as the reference temperature.

noted, however, that the terminal slopes cannot be observed probably due to the higher polydispersity of these polymers. When the NP_7 concentration increases to 7 and 10 wt% (Fig. 4b), the 'weak gel-like' regime is observed, where a plateau is observed at low frequencies for both moduli. The crossing of the two moduli occurs at higher frequencies with increasing nanoparticle concentration similarly to the PEO 35 k / NP_7 system. For a concentration of ~ 15 wt% in NP_7, the 'strong gel-like' regime is observed (not shown). For the PEO300k / NP_7 nanocomposites, it is clear that the hybrid with 2 wt% NP_7 is in the 'polymer-like' regime (Fig. 4c) whereas the hybrid with 5 wt% NP_7 is in the 'weak gel-like' one (Fig. 4d). The 'strong gel-like' regime is reached at 10 wt% NP_7. Therefore, all three regimes can be identified even for higher molecular weights with the transition between the 'polymer-like' and the 'weak gel-like' regime shifting towards higher concentrations with increasing molecular weight.

Two more observations should be emphasized in the data of Fig. 4, especially, for the systems that show the 'polymer-like' behavior (Fig. 4a and 4c). The addition of a small number of nanoparticles shifts the longest relaxation time of the hybrids towards higher frequencies, which indicates faster dynamics in the nanocomposites. Even more, in the case of PEO300k and its nanocomposite with 2 wt% NP_7, where the beginning of the rubbery plateau can be observed in the high frequency side of the data (Fig. 4c), its value for the nanocomposite is significantly lower than that of the neat PEO300k; since the rubbery plateau is inversely proportional to the entanglement molecular weight, M_e, this finding signifies an increase of M_e and, therefore, a dilution of the polymer entanglement network.

The effects of the nanoparticle size on the rheological response of the nanohybrids was investigated as well. Fig. 5 shows the frequency dependence of the storage, G', and the loss, G″, moduli for PEO100k nanohybrids that contain NP_{67} nanoparticles at different concentrations following the time–temperature superposition principle at T_{ref} = 80 °C as the reference temperature. These nanoparticles have a radius of 67 nm, i.e., they are much larger than all three polymers investigated so far. In comparison with the nanohybrids with small NP_7 nanoparticles, the three concentration regimes are shifted towards higher concentrations. For all concentrations up to 25 wt% in NP_{67}, the behavior resembles that of the neat polymer. It is noted that for the nanocomposites with NP_7, this regime was limited to 4 wt%. These results can be understood if one takes into account that NP_{67} are much larger than NP_7 and, therefore, at the same concentration, their number density is much smaller and, thus, they are more 'dilute' within the polymer matrix. Along the same lines, the concentration where the percolated network of nanoparticles is formed is significantly higher (~40 wt%) in comparison with the respective one of NP_7, that was found to be ~ 15 wt%.

Fig. 6 shows the complex (dynamic) viscosity data of all series of nanohybrids composed of the three different PEOs, PEO35k, PEO100k and PEO300k and both NP_7 and NP_{67} nanoparticles. In all series, the complex viscosity data for lower concentrations do not show any frequency dependence. The most important result, however, is the significant decrease of the viscosity with the addition of a small concentration of nanoparticles, which reaches even 1–2 orders of magnitude, contrary to what one would, in general, anticipate. This decrease is observed for all three molecular weights and for both nanoparticle sizes. It is emphasized that in all nanocomposite systems where this reduction was observed, the nanohybrids exhibited the 'polymer-like' regime and the zero-shear viscosity could be clearly determined. Moreover, the range of concentrations where the decrease of the viscosity is observed depends on the molecular weight of the polymer matrix with this range decreasing as the polymer molecular weight decreases. For nanohybrids with higher nanoparticle concentrations, the viscosity increases with further increasing the nanoparticle concentration. Moreover, for higher

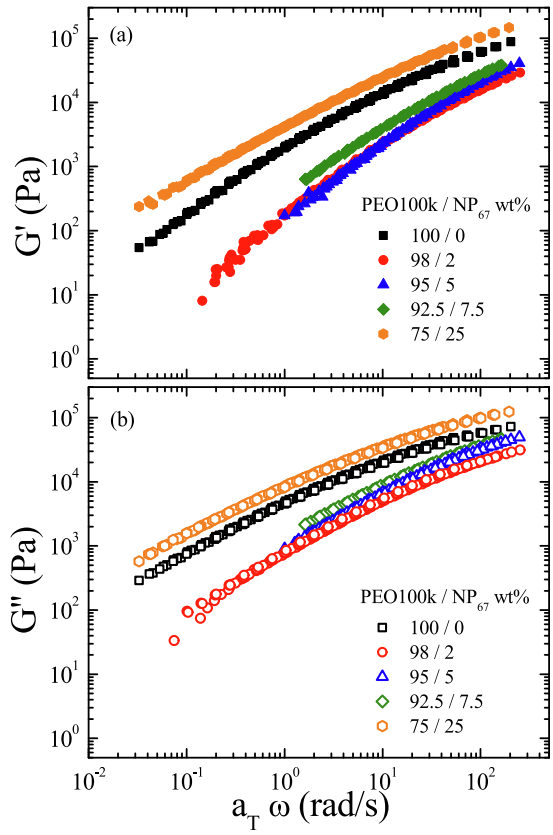

Fig. 5. Frequency dependence of (a) the storage modulus, G', and (b) the loss modulus, G″, for PEO100k / NP_{67} nanohybrids in different compositions measured over a temperature range 60 to 120 °C. The data are shifted along the horizontal axis keeping 80 °C as the reference temperature.

nanoparticle content, with the hybrids entering the 'weak gel-like' regime, an ω^{-1} dependence develops at low frequencies as well.

A decrease in the viscosity of the nanocomposites has been observed in earlier works as well; however, in all those cases that decrease [40,61,62,64–69] was never as significant as in the present work. In certain cases, when the utilized nanoparticles had a smaller size than the size of the polymer chains, the behavior was attributed to a plasticizing effect that was induced by the small nanoparticles, which would act like solvent molecules. Nevertheless, such an explanation is by no means valid in the present case since the decrease in the viscosity is observed for both nanoparticle sizes and, even more, it is of similar magnitude when the NP_{67} nanoparticles are utilized, which are larger than all three higher molecular weight PEOs; as shown in Table 1, the radii of gyration of these polymers range from 9 nm for PEO35k to 46 nm for PEO300k.

Alternatively, the viscosity reduction may be related to an effective dilution of the entanglement network of the polymer by the nanoparticles. Since PEO35k, PEO100k and PEO300k are well-entangled, in order to examine if such an assumption is correct, the behavior of nanohybrids containing polymers of much lower molecular weights should be investigated. Therefore, additional series of nanohybrids have been developed with PEO3.35k and PEO8k, which are unentangled or only marginally entangled; both NP_7 and NP_{67} nanoparticles were used. Fig. 7a shows the viscosity data of the neat PEO3.35k and its nanocomposite with 2 wt% NP_{67}. An unambiguous increase of the viscosity η by the addition of the NP_{67} nanoparticles is observed. It is noted that for nanohybrids of similar composition using PEO35k, PEO100k and PEO300k, as shown in Fig. 6, the viscosity shows a significant decrease. An increase of the viscosity is also observed for nanohybrids of PEO3.35k with lower NP_{67} concentrations as well as for nanohybrids containing the small NP_7 nanoparticles. Further increase of the nanoparticle content leads to the observation of the 'weak gel-like' regime for this system as well as shown in Fig. 7b where the frequency dependence of the storage and the loss moduli are shown for a 5 wt% PEO3.35k / NP_{67} nanocomposite. In this case, a crossing of the G' and the G″ data is observed at intermediate frequencies. Note that, at the particular concentration of 5 wt%, all other hybrids that contain PEOs with higher molecular weights and NP_{67} nanoparticles exhibited a 'polymer-like' behavior. For the PEO3.35k / NP_7 nanohybrids, the viscosity increases for all concentrations, whereas the 'weak gel-like' regime is observed for concentrations as low as 2 wt% NP_7, as shown in Figure S6 of the Supplementary Material. Therefore, the frequency that the crossing occurs depends both on the polymer molecular weight and the nanoparticle size as well as on the nanohybrid concentration.

For the nanocomposites that contain PEO8k, which is marginally entangled, there is no significant dependence of the complex viscosity on the nanoparticle concentration when the hybrids are in the 'polymer-like' regime; for the PEO8k / NP_{67} series, the 'weak gel-like' regime is observed for concentrations higher than 15 wt% in NP_{67}, as shown in Figure S7 of the Supplementary Material.

The overall dependence of the viscosity on the nanoparticle content is shown in Fig. 8, normalized to the corresponding viscosity of the respective neat polymer, for all polymers and nanoparticles used. For the systems exhibiting 'polymer-like' behavior, the zero shear viscosities are shown, which were obtained by the analysis of the frequency-dependent complex viscosity data utilizing the Cross model [74]. For the systems that exhibited a 'weak gel-like' response, the viscosity values are shown at the frequencies where the crossing of G' and G″ occurs.

A non-monotonic behavior is observed for the nanohybrids composed of the three higher molecular weight polymers (PEO 35k, PEO100k, PEO300k), which are highly entangled, for both small and large nanoparticles; the viscosity values initially decrease with the addition of a small concentration of nanoparticles, they reach a minimum and, then, increase with further increasing of the nanoadditive content. For the PEO8k, which is a marginally entangled polymer, the viscosity remains relatively constant for a large range of concentrations and eventually crosses over to the 'weak gel-like' regime (between 10 and 15 wt%). Finally, for PEO3.35k, which is the lowest molecular weight unentangled polymer studied, the viscosity shows an immediate increase with the addition of even a small concentration of nanoparticles; the latter behavior is observed for both small and large nanoparticles. Overall, the significant decrease of the viscosity in these polymer nanocomposites by almost two orders of magnitude is a remarkable finding that has not been so clearly observed before. Moreover, the data present a clear evidence that this decrease is related to an effective dilution of the entanglement network of the polymer chains due to the presence of the nanoparticles.

A reduction of the viscosity of nanocomposite systems has been observed before in several different works; in all cases, however, the reduction was not to such a degree as in this present work. Some of the most well discussed systems are the ones reported from Mackay, et. al. [61,65] where polystyrene nanoparticles were dispersed within a linear polystyrene matrix, thus creating an ideal system. A decrease of the zero shear viscosity by ~ 60 % was observed [61], which was attributed to an increase of free volume, whereas increase of the molecular weight of the polymer matrix led to a decrease of the zero shear viscosity by ~ 80 % [65]. In a different work with similar very small (~2 nm) polystyrene nanoparticles within linear polystyrene matrices of different molecular weights, Chen, et al. [67] showed a viscosity reduction that is amplified with an increase in molecular weight that reached ~ 60 % viscosity reduction; in that case, the effect was attributed to a reduced friction experienced by polymer segment blobs, which have similar size with the nanoparticles. When Au nanoparticles coated with PEO chains were dispersed within an entangled PEO matrix (like our PEO35k), Senses et. al. [70] showed only a ~ 60 % reduction with the small (~3.5 nm) nanoparticles and a reinforcement with larger (~20 nm) nanoparticles; in that work, disentanglement of the polymer chains rather than

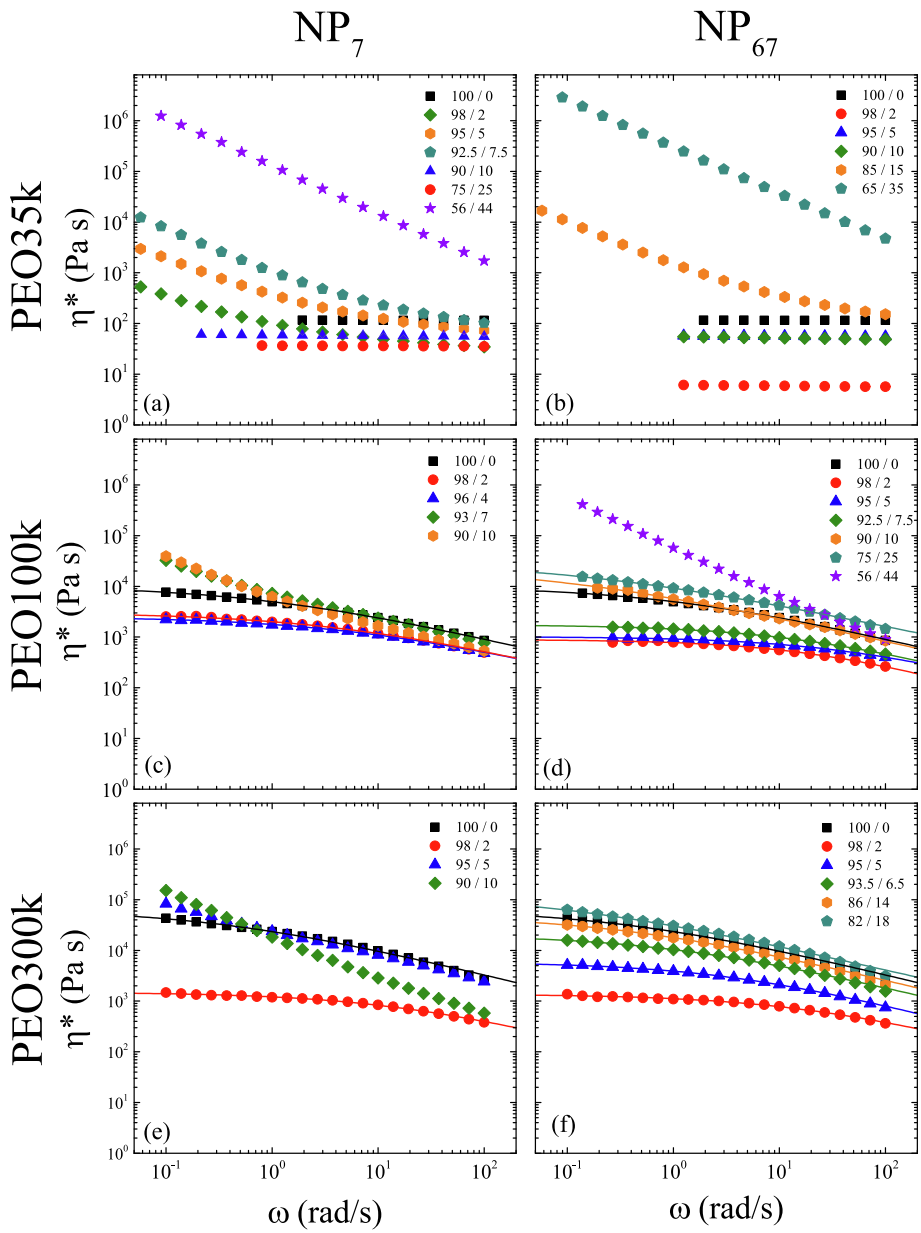

Fig. 6. Frequency dependence of the complex viscosity, η*, for the different series of PEO/SiO_2 nanocomposites containing three PEO polymers, PEO35k (a, b), PEO100k (c, d) and PEO300k (e, f) and two different nanoparticles, NP_7 (a, c, e) and NP_{67} (b, d, f). The concentrations are indicated as weight %. The lines in (c-f) are fittings utilizing the Cross model.

plasticization or free volume increase was discussed. Moreover, in a nanocomposites with zirconia nanoparticles with grafted poly(dimethyl siloxane), PDMS, chains dispersed in PDMS of different molecular weights, viscosity reductions by ~ 40 % were observed and only for the system with the higher molecular weight (77 k) of the matrix attributed to plasticization effects. Therefore, the present work is the only one that shows such a significant reduction of the viscosity (by almost two orders of magnitude) observed for nanoparticles both smaller and larger than the size of the polymer.

Note that, for these nanohybrids, a decrease in the polymer crystallinity had been observed, which was attributed to a synergetic effect of the confinement of polymer chains that the nanoparticles impose and on the adsorption capacity of the nanoadditive surfaces due to the favorable interactions between the nanoparticle surfaces and the polymer segments [29]. However, it is emphasized that the range of nanoparticle concentrations, where the decrease of the viscosity is observed, is very much smaller than that associated with the onset of the decrease of the degree of crystallinity, as shown in the Differential Scanning Calorimetry measurements of Figure S8 of the Supplementary Material for the PEO100k/NP_7 and PEO100/NP_{67} nanocomposites. On the contrary, in the specific composition regime, a slight increase of the degree of crystallinity is observed which could be attributed to the dilution of the entanglements in conjunction with the nucleation ability of the nanoparticles.

4. Conclusions

The linear rheology of a series of poly(ethylene oxide) / silica, (PEO/SiO_2) nanocomposites was investigated. The nanohybrids covered a wide range of compositions utilizing nanoparticles of two different radii dispersed in PEOs of different molecular weights from unentangled to well-entangled. This has allowed to systematically investigate the effects of the size and concentration of the nanoparticles as well as the effects of the molecular weight of the polymer matrix, on the linear viscoelastic

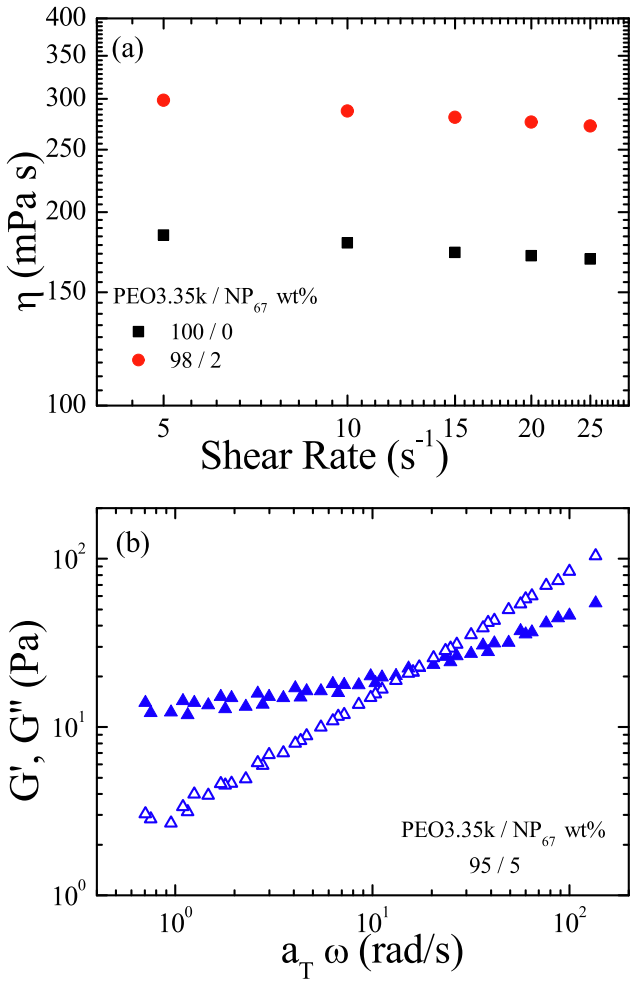

Fig. 7. Shear rate dependence of the viscosity for PEO3.35 k and PEO3.35k / NP_{67} with 2 wt% of nanoparticles. (b) Frequency dependence of the storage, G', and the loss modulus, G″, for PEO3.35k / NP_{67} with 5 wt% nanoparticles. The measurements were obtained over a temperature range 60 to 100 °C and are presented following time–temperature superposition principle by shifting the data on the x-axis utilizing 80 °C as the reference temperature.

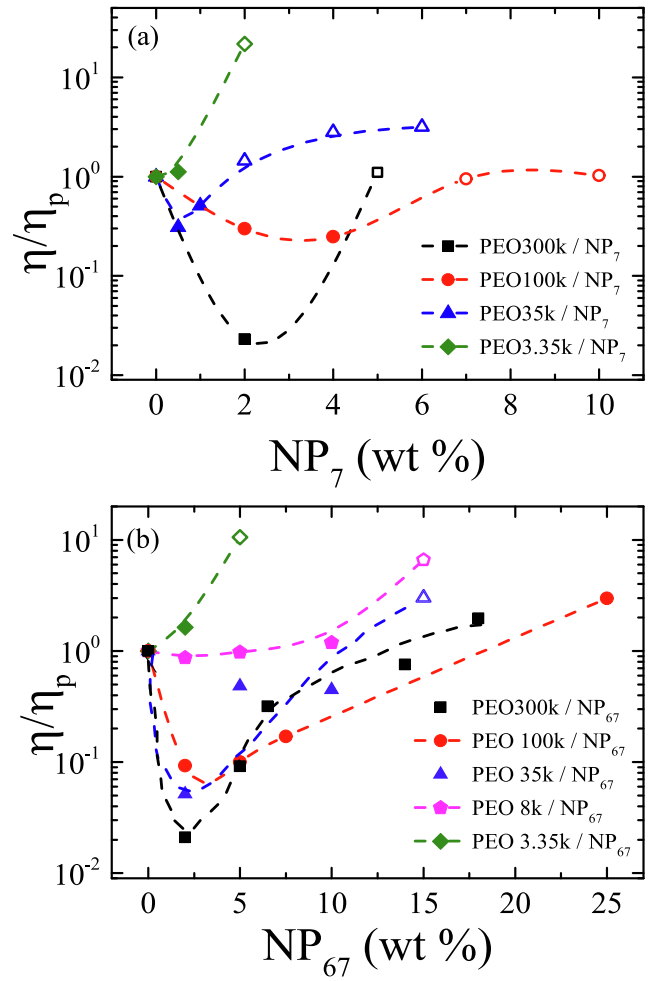

Fig. 8. Dependence of the viscosity of the nanocomposites, normalized to the value of the respective polymers, on the nanoparticle weight fraction for nanocomposites containing (a) NP_7 and (b) NP_{67} nanoparticles at 80 °C. The solid points are the zero shear viscosity obtained by fitting the complex viscosity data with the Cross model whereas the open symbols are the viscosity values at the frequencies where the crossing of G' and G″ occurs for the systems that exhibited the 'weak gel-like' response. The lines through the points are guide to the eye. In both (a) and (b) the error in the viscosity value is smaller than the size of the points.

properties of the nanocomposites. In all systems, three different concentration regimes were observed with different rheological response; a 'polymer-like' regime, where the behavior resembles the one of the respective polymer matrices, a 'weak gel-like' regime at intermediate nanoparticle content, where enhanced polymer / particle interactions modify the behavior due to an adsorption/desorption mechanism of the chains onto the nanoparticles, which would slow down the polymer motion, and a 'strong gel-like' one at the highest concentrations, when percolated networks of nanoparticles are formed within the polymer matrices. The borders of the three regimes are determined by both the size of the nanoparticles and the polymer molecular weight.

Of particular interest is the 'polymer-like' regime where a significant reduction of the melt viscosity was observed, for both nanoparticle sizes but only for the three highly-entangled systems. On the contrary, the nanocomposites with the lower molecular weight unentangled polymer exhibited a continuous increase of the viscosity, whereas for the marginally-entangled polymer, apparently any possible reduction is counter-balanced by the reinforcing effect of the nanoparticles. At the same time, although the viscosity reduction is clearly observed for both nanoparticles, the quantification of the effect of the nanoparticle size on the viscosity reduction is complicated by the appearance of the gel-like behavior as concentration increases. All these facts lead to the conclusion that it is the effective dilution of the entanglement network of the polymer chains because of the presence of the nanoparticles, even at small concentrations, that is at the origin of this behavior.

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CRediT authorship contribution statement

Thaleia-Michaela Chatzaki: Writing – original draft, Methodology, Investigation, Validation. **Sokratis Kogchylakis:** Investigation, Methodology, Validation. **Dimitris Vlassopoulos:** Writing – review & editing, Resources. **Spiros H. Anastasiadis:** Writing – review & editing, Funding acquisition, Project administration, Resources. **Kiriaki Chrissopoulou:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.112903.

References

[1] D. Schmidt, D. Shah, E.P. Giannelis, New advances in polymer / layered silicate nanocomposites, Curr. Opin. Sol. Stat. Mater. Sci. 6 (2002) 205–212.
[2] A. Bansal, H. Yang, C. Li, K. Cho, B.C. Benicewicz, S.K. Kumar, L.S. Schadler, Quantitative equivalence between polymer nanocomposites and thin polymer films, Nature Mater. 4 (2005) 693–698.
[3] M. Moniruzzaman, K.I. Winey, Polymer nanocomposites containing carbon nanotubes, Macromolecules 39 (2006) 5194–5205.
[4] Y.Q. Rao, J.M. Pochan, Mechanics of Polymer−Clay nanocomposites, Macromolecules 40 (2007) 290–296.
[5] P. Akcora, H. Liu, S.K. Kumar, J. Moll, Y. Li, B.C. Benicewicz, L.S. Schadler, D. Acehan, A.Z. Panagiotopoulos, V. Pryamitsyn, V. Ganesan, J. Ilavsky, P. Thiyagarajan, R.H. Colby, J.F. Douglas, Anisotropic self-assembly of spherical polymer-grafted nanoparticles, Nature Mater. 8 (2009) 354–359.
[6] J. Jancar, J.F. Douglas, F.W. Starr, S.K. Kumar, P. Cassagnau, A.J. Lesser, S. S. Sternstein, M.J. Buehler, Current issues in research on structure–property relationships in polymer nanocomposites, Polymer 51 (2010) 3321–3343.
[7] V. Mittal, Functional polymer nanocomposites with graphene: a review, Macrom. Mater. Eng. 299 (2014) 906–931.
[8] A. Karatrantos, N. Clarke, M. Kröger, Modeling of polymer structure and conformations in polymer nanocomposites from atomistic to mesoscale: a review, Polym. Rev. 56 (2016) 385–428.
[9] D. Feldman, Polymer nanocomposites in medicine, J. Macromol. Sci. Part a: Pur. Appl. Chem. 53 (2016) 55–62.
[10] A.K. Naskar, J.K. Keum, R.G. Boeman, Polymer matrix nanocomposites for automotive structural components, Nat. Nanotech. 11 (2016) 1026–1030.
[11] E. Glynos, L. Papoutsakis, W. Pan, E.P. Giannelis, A.D. Nega, E. Mygiakis, G. Sakellariou, S.H. Anastasiadis, Nanostructured polymer particles as additives for high conductivity, high modulus solid polymer electrolytes, Macromolecules 50 (2017) 4699–4706.
[12] I.A. Kinloch, J. Suhr, Composites with carbon nanotubes and graphene: an outlook, Science 362 (2018) 547–553.
[13] S. Ganguly, P. Bhawal, R. Ravindren, N.C. Das, Polymer nanocomposites for electromagnetic interference shielding: a review, J. Nanosci. Nanotech. 18 (2018) 7641–7669.
[14] I. Pastoriza-Santos, C. Kinnear, J. Perez-Juste, P. Mulvaney, L.M. Liz-Marzan, Plasmonic polymer nanocomposites, Nat. Rev. Mat. 3 (2018) 375–391.
[15] M. Abbas, M. Buntinx, W. Deferme, R. Peeters, (Bio)polymer/ZnO Nanocomposites for Packaging Applications; A Review of Gas Barrier and Mechanical Properties, Nanomaterials 9 (2019) 1494 (1-14).
[16] V. Melinte, L. Stroea, A.L. Chibac-Scutaru, Polymer Nanocomposites for Photocatalytic Applications, Catalysts 9 (2019) 986 (1-28).
[17] D. Feldman, Polyurethane and polyurethane nanocomposites: recent contributions to medicine, Biointerf. Res. Appl. Chem. 11 (2021) 8179–8189.
[18] F. Krasanakis, T.-M. Chatzaki, K. Chrissopoulou, S.H. Anastasiadis, Modifying flexible polymer films towards superhydrophobicity and superoleophobicity by utilizing water-based nanohybrid coatings, Nanoscale 15 (2023) 6984–6998.
[19] M.M. Elmahdy, K. Chrissopoulou, A. Afratis, G. Floudas, S.H. Anastasiadis, Effect of confinement on polymer segmental motion and ion mobility in PEO/LAYERED silicate nanocomposites, Macromolecules 39 (2006) 5170–5173.
[20] K. Chrissopoulou, K.S. Andrikopoulos, S. Fotiadou, S. Bollas, C. Karageorgaki, D. Christofilos, G.A. Voyiatzis, S.H. Anastasiadis, Crystallinity and chain conformation in PEO/LAYERED silicate nanocomposites, Macromolecules 44 (2011) 9710–9722.
[21] J.M. Carr, D.S. Langhe, M.T. Ponting, A. Hiltner, E. Baer, Confined crystallization in polymer nanolayered films: a review, J. Mater. Res. 27 (2012) 1326–1350.
[22] F. Barroso-Bujans, S. Cerveny, Á. Alegría, J. Colmenero, Chain length effects on the dynamics of poly(ethylene oxide) confined in graphite oxide: a broadband dielectric spectroscopy study, Macromolecules 46 (2013) 7932–7939.
[23] G.G. Vogiatzis, D.N. Theodorou, Local segmental dynamics and stresses in Polystyrene–C60 mixtures, Macromolecules 47 (2014) 387–404.
[24] K. Androulaki, K. Chrissopoulou, D. Prevosto, M. Labardi, S.H. Anastasiadis, Dynamics of hyperbranched polymers under confinement: a dielectric relaxation study, ACS Appl. Mater. Inter. 7 (2015) 12387–12398.
[25] R.M. Michell, A.J. Müller, Confined crystallization of polymeric materials, Prog. Polym. Sci. 54–55 (2016) 183–213.
[26] W. Zhao, Y. Su, A.J. Müller, X. Gao, D. Wang, Direct relationship between interfacial microstructure and confined crystallization in poly(ethylene oxide) / silica composites: the study of polymer molecular weight effects, J. Polym. Sci. Part b: Polym. Phys. 55 (2017) 1608–1616.
[27] S. Bollas, K. Chrissopoulou, K. Andrikopoulos, G.A. Voyiatzis, S.H. Anastasiadis, Polymer conformation under confinement, Polymers 9 (2017) 73.
[28] A.N. Rissanou, H. Papananou, V.S. Petrakis, M. Doxastakis, K.S. Andrikopoulos, G. A. Voyiatzis, K. Chrissopoulou, V. Harmandaris, S.H. Anastasiadis, Structural and conformational properties of poly(ethylene oxide) / silica nanocomposites: effect of confinement, Macromolecules 50 (2017) 6273–6284.
[29] H. Papananou, E. Perivolari, K. Chrissopoulou, S.H. Anastasiadis, Tuning polymer crystallinity via the appropriate selection of inorganic nanoadditives, Polymer 157 (2018) 111–121.
[30] P. Bačová, W. Li, A.F. Behbahani, C. Burkhart, P. Polińska, M. Doxastakis, V. Harmandaris, Coupling between polymer conformations and dynamics near amorphous silica surfaces: a direct insight from atomistic simulations, Nanomaterials 11 (2021) 2075.
[31] K. Androulaki, K. Chrissopoulou, M. Labardi, S.H. Anastasiadis, Effect of interfacial interactions on static and dynamic behavior of hyperbranched polymers: comparison between different layered nanoadditives, Polymer 222 (2021) 123646.
[32] K. Chrissopoulou, K. Androulaki, M. Labardi, S.H. Anastasiadis, Static and dynamic behavior of polymer/graphite oxide nanocomposites before and after thermal reduction, Polymers 13 (2021) 1008.
[33] A.J. Power, H. Papananou, A.N. Rissanou, M. Labardi, K. Chrissopoulou, V. Harmandaris, S.H. Anastasiadis, Dynamics of polymer chains in poly(ethylene oxide) / silica nanocomposites via a combined computational and experimental approach, J. Chem. Phys. B 126 (2022) 7745–7760.
[34] A.A. Teran, M.H. Tang, S.A. Mullin, N.P. Balsara, Effect of molecular weight on conductivity of polymer electrolytes, Solid State Ionics 203 (2011) 18–21.
[35] X. Fan, Z. Hu, G. Wang, Synthesis and unimolecular micelles of amphiphilic copolymer with dendritic poly(l-lactide) core and poly(ethylene oxide) shell for drug delivery, RSC Adv. 5 (2015) 100816–100823.
[36] Z. Tang, C. He, H. Tian, J. Ding, B.S. Hsiao, B. Chu, X. Chen, Polymeric nanostructured materials for biomedical applications, Prog. Polym. Sci. 60 (2016) 86–128.
[37] E. Glynos, P. Petropoulou, E. Mygiakis, A.D. Nega, W.Y. Pan, L. Papoutsakis, E. P. Giannelis, G. Sakellariou, S.H. Anastasiadis, Leveraging molecular architecture to design new, all-polymer solid electrolytes with simultaneous enhancement in modulus and ionic conductivity, Macromolecules 51 (2018) 2542–2550.
[38] F. Croce, G.B. Appetecchi, L. Persi, B. Scrosati, Nanocomposite polymer electrolytes for lithium batteries, Nature 394 (1998) 456–459.
[39] J. Zhang, H. Han, S. Wu, S. Xu, Y. Yang, C. Zhou, X. Zhao, Conductive carbon nanoparticles hybrid PEO/P(VDF-HFP)/SiO2 nanocomposite polymer electrolyte type dye sensitized solar cells, Solid State Ionics 178 (2007) 1595–1601.
[40] A. Kelarakis, E.P. Giannelis, Crystallization and unusual rheological behavior in poly(ethylene oxide)/clay nanocomposites, Polymer 52 (2011) 2221–2227.
[41] S. Srivastava, J.L. Schaefer, Z. Yang, Z. Tu, L.A. Archer, 25th anniversary article: polymer-particle composites: phase stability and applications in electrochemical energy storage, Adv. Mater. 26 (2014) 201–234.
[42] O.E. Philippova, S.I. Kuchanov, I.N. Topchieva, V.A. Kabanov, Hydrogen bonds in dilute solutions of poly(ethylene glycol), Macromolecules 18 (1985) 1628–1633.
[43] Y. Cao, Y. Guan, J. Du, J. Luo, Y. Peng, C.W. Yip, A.S.C. Chan, Hydrogen-bonded polymer network—poly(ethylene glycol) complexes with shape memory effect, J. Mater. Chem. 12 (2002) 2957–2960.
[44] D. Barbier, D. Brown, A.-C. Grillet, S. Neyertz, Interface between end-functionalized PEO oligomers and a silica nanoparticle studied by molecular dynamics simulations, Macromolecules 37 (2004) 4695–4710.
[45] E.N. Skountzos, D.G. Tsalikis, P.S. Stephanou, V.G. Mavrantzas, Individual contributions of adsorbed and free chains to microscopic dynamics of unentangled poly(ethylene glycol)/silica nanocomposite melts and the important role of end groups: theory and simulation, Macromolecules 54 (2021) 4470–4487.
[46] Q. Zhang, L.A. Archer, Poly(ethylene oxide)/silica nanocomposites: structure and rheology, Langmuir 18 (2002) 10435–10442.
[47] S.K. Kumar, R. Krishnamoorti, Nanocomposites: structure, phase behavior, and properties, Annu. Rev. Chem. Biomol. Eng. 1 (2010) 37–58.
[48] G.P. Baeza, C. Dessi, S. Costanzo, D. Zhao, S. Gong, A. Alegria, R.H. Colby, M. Rubinstein, D. Vlassopoulos, S.K. Kumar, Network dynamics in nanofilled polymers, Nat. Commun. 7 (2016) 11368.
[49] Y. Song, Q. Zhang, Size-dependent linear rheology of silica filled poly(2-vinylpyridine), Polymer 130 (2017) 74–78.

[50] Z.Y. Zhu, T. Thompson, S.Q. Wang, E.D. von Meerwall, Investigating linear and nonlinear viscoelastic behavior using model silica-particle-filled polybutadiene, Macromolecules 38 (2005) 8852–8866.
[51] P. Akcora, S.K. Kumar, J. Moll, S. Lewis, L.S. Schadler, Y. Li, B.C. Benicewicz, A. Sandy, S. Narayanan, J. Ilansky, P. Thiyagarajan, R.H. Colby, J.F. Douglas, Gel-like mechanical reinforcement in polymer nanocomposite melts, Macromolecules 43 (2010) 1003–1010.
[52] B.J. Anderson, C.F. Zukoski, Rheology and microstructure of an unentangled polymer nanocomposite melt, Macromolecules 41 (2008) 9326–9334.
[53] R. Mangal, S. Srivastava, S. Narayanan, L.A. Archer, Size-dependent particle dynamics in entangled polymer nanocomposites, Langmuir 32 (2016) 596–603.
[54] E. Senses, S. Narayanan, Y. Mao, A. Faraone, Nanoscale particle motion in attractive polymer nanocomposites, Phys. Rev. Lett. 119 (2017) 237801–237806.
[55] D. Long, F. Lequeux, Heterogeneous dynamics at the glass transition in van der waals liquids, in the bulk and in thin films, Eur. Phys. J. E 4 (2001) 371–387.
[56] J. Berriot, H. Montes, F. Lequeux, D. Long, P. Sotta, Evidence for the shift of the glass transition near the particles in silica-filled elastomers, Macromolecules 35 (2002) 9756–9762.
[57] M. Krutyeva, A. Wischnewski, M. Monkenbusch, L. Wilner, J. Maiz, C. Mijangos, A. Arbe, J. Colmenero, A. Radulescu, O. Holderer, M. Ohl, D. Richter, Effect of nanoconfinement on polymer dynamics: surface layers and interphases, Phys. Rev. Lett. 110 (2013) 108303.
[58] Q. Chen, S. Gong, J. Moll, D. Zhao, S.K. Kumar, R.H. Colby, ACS Macro Letters 4 (2015) 398–402.
[59] H. Reda, A. Chazirakis, A.F. Behbahani, N. Savva, V. Harmandaris, Revealing the role of chain conformations on the origin of the mechanical reinforcement in glassy polymer nanocomposites, Nano Letters (2023) in press.
[60] V. Guzeyev, M. Rafikov, Y. Malinskii, The effect of filler dispersity of the melt viscosity of polyvinyl chloride (PVC), Polym. Sci. USSR 5 (1975) 804.
[61] M.E. Mackay, T.T. Dao, A. Tuteja, D.L. Ho, B. van Horn, H.C. Kim, C.J. Hawker, Nanoscale effects leading to non-einstein-like decrease in viscosity, Nat. Mater. 2 (2003) 762–766.
[62] K. Nusser, G.J. Schneider, W. Pyckhout-Hintzen, D. Richter, Viscosity decrease and reinforcement in polymer-silsequioxane composites, Macromolecules 44 (2011) 7820–7830.
[63] J.T. Kalathi, G.S. Grest, S.K. Kumar, Universal viscosity behavior of polymer nanocomposites, Phys. Rev. Lett. 109 (2012) 198301.
[64] R. Mangal, S. Srivastava, L.A. Archer, Phase stability and dynamics of entangled polymer–nanoparticle composites, Nat. Commun. 6 (2015) 7198.
[65] A. Tuteja, M.E. Mackay, C.J. Hawker, B. van Horne, Effect of ideal, organic nanoparticles on the flow properties of linear polymers: non-einstein-like behavior, Macromolecules 38 (2005) 8000–8011.
[66] R.G. Schmidt, G.V. Gordon, C.A. Dreiss, T. Cosgrove, V.J. Krukonis, K. Williams, P. M. Wetmore, Critical size ratio for viscosity reduction in Poly(dimethylsiloxane)–Polysilicate nanocomposites, Macromolecules 43 (2010) 10143–10151.
[67] T. Chen, H.-Y. Zhao, R. Shi, W.-F. Lin, X.-M. Jia, H.-J. Qian, Z.-Y. Lu, X.-X. Zhang, Y.-K. Li, Z.-Y. Sun, An unexpected N-dependence in the viscosity reduction in all-polymer nanocomposite, Nat. Comm. 10 (2019) 5552.
[68] E. Senses, S.M. Ansar, C.L. Kitchens, Y. Mao, S. Narayanan, B. Natarajan, A. Faraone, Small particle driven chain disentanglements in polymer nanocomposites, Phys. Rev. Lett. 118 (2017) 147801.
[69] M. Giovino, E. Buenning, A. Jimenez, S.K. Kumar, L. Schadler, Polymer grafted nanoparticle viscosity modifiers, Macromol. Chem. Phys. 220 (2019) 1800543.
[70] E. Senses, C.L. Kitchens, A. Faraone, Viscosity reduction in polymer nanocomposites: insights from dynamic neutron and X-ray scattering, J. Polym. Sci. 60 (2022) 1130–1150.
[71] L. Peng, W. Qisui, L. Xi, Z. Chaocan, Investigation of the states of water and OH groups on the surface of silica, colloids and surfaces a: physicochem, Eng. Aspects 334 (2009) 112–115.
[72] M.M. Cross, Rheology of non-newtonian fluids: a new flow equation for pseudoplastic systems, J. Colloid Sci. 20 (1965) 417–437.
[73] W.W. Graessley, S.F. Edwards, Entanglement interactions in polymers and the chain contour concentration, Polymer 22 (1981) 1329–1334.
[74] H.H. Winter, M. Mours, Rheology of polymers near liquid-solid transitions, Adv. Polym. Sci. 134 (1997) 165–234.
[75] W. Cui, W. You, Z. Sun, W. Yu, Decoupled polymer dynamics in weakly attractive poly(methyl methacrylate)/silica nanocomposites, Macromolecules 54 (2021) 5484–5497.
[76] S. Wu, Entanglement, friction and free volume between dissimilar chains in compatible polymer blends, J. Polym. Sci., Part b: Polym. Phys. 25 (1987) 2511–2529.
[77] L.J. Fetters, D.J. Lohse, D. Richter, T.A. Witten, A. Zirkel, Connection between polymer molecular weight, density, chain dimensions, and melt viscoelastic properties, Macromolecules 27 (1994) 4639–4647.
[78] L.J. Fetters, D.J. Lohse, R.H. Colby, in: Physical Properties of Polymers Handbook, second ed., Springer Science, New York, 2007, pp. 447–454.